

- 23** 10 cm^3 of a gaseous hydrocarbon, X, burns completely with exactly 75 cm^3 of oxygen to produce 50 cm^3 of carbon dioxide. All volumes are measured at the same temperature and pressure.

Which statements about the hydrocarbon are correct?

- 1 X may be a cycloalkane.
- 2 X may have a branched chain.
- 3 X may decolourise aqueous bromine in the dark.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only